ERP Data Migration & Revenue Integrity Analysis

1. Executive Summary

This project simulates a real-world ERP data migration from a legacy CRM system (HubSpot/Kaggle-style dataset) to a unified ERP environment (Dynamics 365 Finance). The primary objective was to ensure data integrity, identify revenue leakage, and establish a clean financial baseline during system transition.

During the analysis, significant inconsistencies were identified in legacy CRM data, including pricing discrepancies, uncontrolled credit exposure, and mismatches across business systems. These issues resulted in unreliable financial reporting and potential revenue loss.

Using SQL-based validation techniques and Power BI visualizations, the project successfully:

- Achieved 100% financial reconciliation between legacy and ERP systems
- Identified 11.2% revenue leakage (~$1.1M)
- Prevented $339K in potential bad debt through credit risk analysis
- Established a clean revenue baseline of $8.88M

This project demonstrates how data analytics plays a critical role in ensuring data reliability, improving financial accuracy, and supporting enterprise-level decision-making during system migrations.

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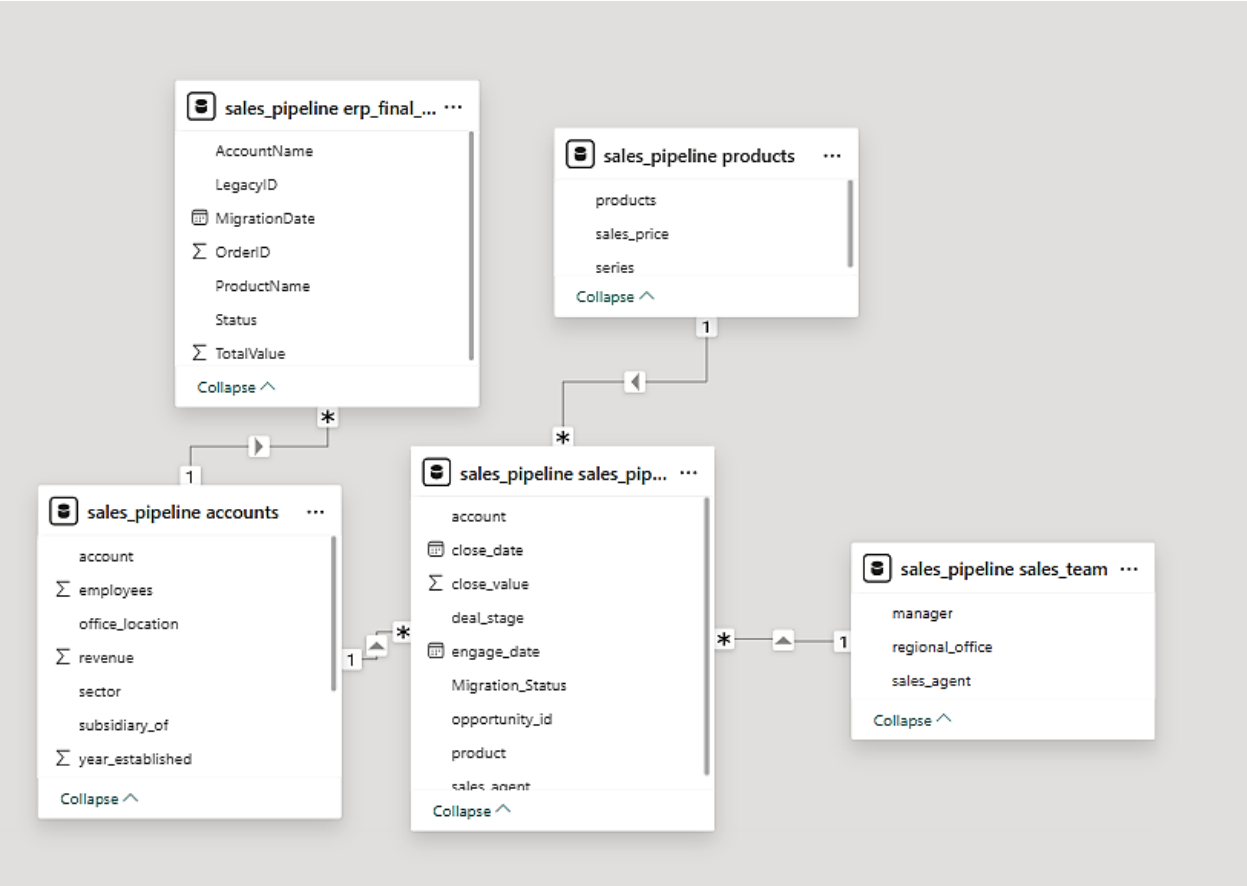
2. Data Structure



3. SQL Query Outputs

```
## Data Validation
-- Missing Accounts (Data Integrity Check)
SELECT
   p.opportunity_id,
   p.account AS Missing_Account
FROM sales_pipeline p
LEFT JOIN accounts a ON p.account = a.account
WHERE a.account IS NULL;
```

Result Grid | Filter Rows:

opportunity_id	Missing_Account
00400B1S	NULL
01EH41WA	NULL
02K37JAK	NULL
03P9VXWG	NULL
03VCDZEE	NULL
04OAV3CV	NULL
04SCU6DK	NULL

Result 1

```
## Price Variance Analysis
SELECT
    p.opportunity_id,
    p.product AS Pipeline_Item,
    pr.products AS Master_Item,
    p.close_value AS Actual_Sold_Price,
    pr.sales_price AS Master_List_Price,
    (p.close_value - pr.sales_price) AS Price_Variance
FROM sales_pipeline p
INNER JOIN products pr ON p.product = pr.products
WHERE p.deal_stage = 'Won';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

opportunity_id	Pipeline_Item	Master_Item	Actual_Sold_Price	Master_List_Price	Price_Variance
0000I7AO	GTX Basic	GTX Basic	505.00	550.00	-45.00
005JS57S	GTX Basic	GTX Basic	539.00	550.00	-11.00
00KY25OA	GTX Basic	GTX Basic	564.00	550.00	14.00
00XBYXIB	GTX Plus Basic	GTX Plus Basic	993.00	1096.00	-103.00
019I751P	GTX Plus Basic	GTX Plus Basic	1180.00	1096.00	84.00
02EC1993	MG Special	MG Special	57.00	55.00	2.00
02G0N1CN	MG Special	MG Special	68.00	55.00	13.00

```
## Discount / Loss Analysis
SELECT
   p.account,
   ROUND(((pr.sales_price - p.close_value) /
pr.sales_price) * 100, 2) AS Discount_Percentage
FROM sales_pipeline p
INNER JOIN products pr ON p.product = pr.products
WHERE p.deal_stage = 'Won';
```

Result Grid | Filter Rows:

account	Discount_Percentage
Rangreen	8.18
Treequote	2.00
Condax	-2.55
Green-Plus	9.40
Statholdings	-7.66
Scottech	-3.64
Xx-holding	-23.64

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Identifies large discount deals (loss > $500) and calculates discount percentage

```
## Price Variance Detection
SELECT
   p.opportunity_id,
   p.account,
   p.product AS Pipeline_Item,
   p.close_value AS Actual_Sold_Price,
   pr.sales_price AS Master_List_Price,
   (p.close_value - pr.sales_price) AS Price_Variance,
   -- Calculate percentage discount for risk assessment
   ROUND(((pr.sales_price - p.close_value) / pr.sales_price) * 100, 2) AS
Discount_Percentage
FROM sales_pipeline p
INNER JOIN products pr ON p.product = pr.products
WHERE p.deal_stage = 'Won'
  AND (p.close_value - pr.sales_price) < -500 -- Focus on losses greater than $500
ORDER BY Price_Variance ASC; -- Show the biggest losses at the top
```

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Result Grid | Filter Rows: | Export: | Wrap Cell Content:

opportunity_id	account	Pipeline_Item	Actual_Sold_Price	Master_List_Price	Price_Variance	Discount_Percentage
OUIK8VX3	Lexiqvolax	GTK 500	23746.00	26768.00	-3022.00	11.29
TB27K4GC	Finjob	GTK 500	24745.00	26768.00	-2023.00	7.56
83JP1K4E	Rantouch	GTK 500	24899.00	26768.00	-1869.00	6.98
K0T5LJ3E	Plexzap	GTK 500	24949.00	26768.00	-1819.00	6.80
IFVL4WL5	Streethex	GTX Plus Pro	3704.00	5482.00	-1778.00	32.43
JXUXBANJ	Y-corporation	GTK 500	25288.00	26768.00	-1480.00	5.53
6K7LDRKR	Stanredtax	GTX Plus Pro	4009.00	5482.00	-1473.00	26.87

Result 4

Validates whether the ERP migration is complete, accurate, and financially consistent

Final Migration Reconciliation Report

```
SELECT
   'Total Legacy Records' AS Metric, COUNT(*) AS Value FROM sales_pipeline
UNION ALL
SELECT
   'Records Flagged for Review', COUNT(*) FROM sales_pipeline WHERE Migration_Status <>
'Ready'
UNION ALL
SELECT
   'Successfully Migrated (Target)', COUNT(*) FROM ERP_Final_SalesOrders
UNION ALL
SELECT
   'Financial Tie-back (Legacy Total Value)', SUM(close_value) FROM sales_pipeline WHERE
Migration_Status = 'Ready'
UNION ALL
SELECT
   'Financial Tie-back (ERP Total Value)', SUM(TotalValue) FROM ERP_Final_SalesOrders;
```

Result Grid | Filter Rows: | Export:

Metric	Value
Total Legacy Records	8800.00
Records Flagged for Review	147.00
Successfully Migrated (Target)	17306.00
Financial Tie-back (Legacy Total Value)	9225633.00
Financial Tie-back (ERP Total Value)	18451266.00

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Total Revenue Leakage & Variance Analysis

```
SELECT
   'Gross Potential (Legacy)' AS Category,
   SUM(close_value) AS Amount
FROM sales_pipeline
WHERE deal_stage = 'Won'

UNION ALL

-- Pricing Leakage (The 147 records from Alpha)
SELECT
   'Pricing Leakage (Quarantined)',
   SUM(close_value)
FROM sales_pipeline
WHERE Migration_Status = 'Review_Required_Price_Gap'

UNION ALL

-- Credit Risk Leakage (The 236 records from Beta)
SELECT
   'Credit Risk Leakage (Blocked)',
   SUM(TotalValue)
FROM ERP_Final_SalesOrders
WHERE Credit_Status = 'CREDIT_HOLD'

UNION ALL

-- Net Realizable Revenue (The Clean "Ready" orders)
SELECT
   'Net Realizable Revenue (Clean)',
   SUM(TotalValue)
FROM ERP_Final_SalesOrders
WHERE Status = 'Ready_to_Invoice'
  AND Credit_Status = 'Approved';
```

Result Grid | Filter Rows: | Export

Category	Amount
Gross Potential (Legacy)	10005534.00
Pricing Leakage (Quarantined)	779901.00
Credit Risk Leakage (Blocked)	339549.00
Net Realizable Revenue (Clean)	18111717.00

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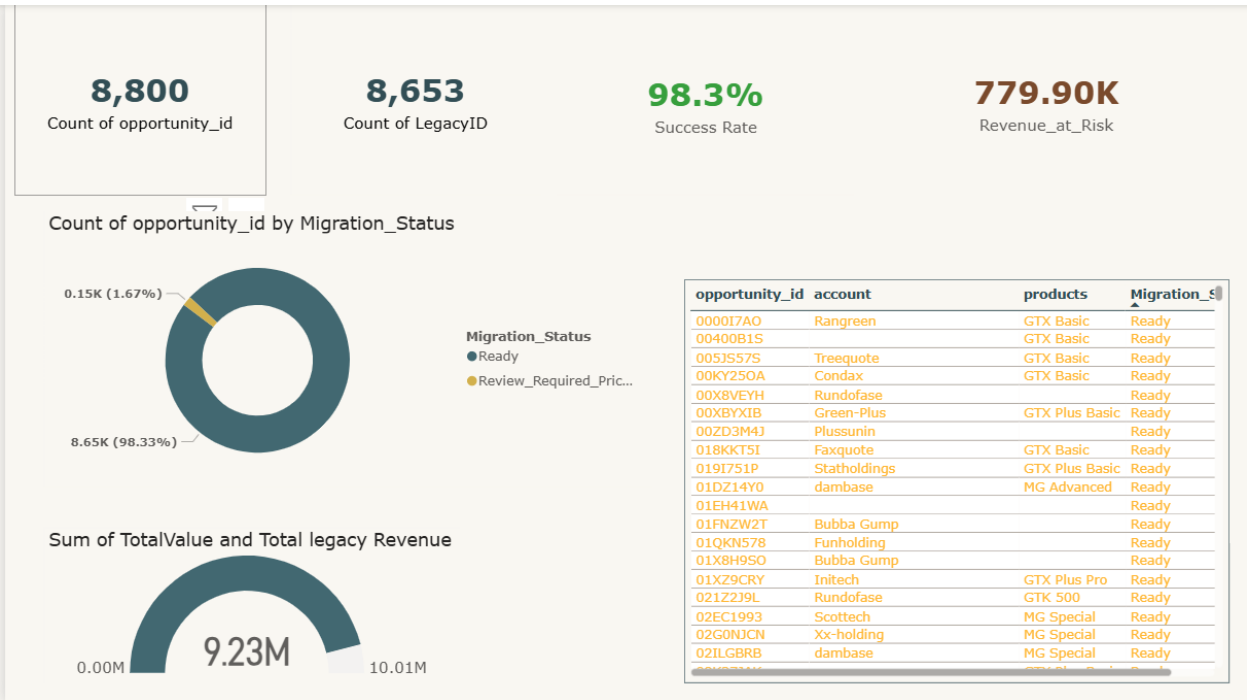
4. Dashboard (Insert Screenshot Below)



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5. Data Wrangling

Data wrangling was a critical step to prepare raw CRM data for analysis and migration**.**

Key activities included:

- Data cleaning: Handling missing values, duplicate records, and inconsistent entries
- Data standardization: Aligning field names and formats between CRM and ERP schemas
- Data validation: Implementing SQL-based checks to identify anomalies and invalid records
- Quarantine logic: Isolating 147 inconsistent records for further review
- Data transformation: Converting business fields (e.g., deal stages, pricing) into ERP-compatible formats

These steps ensured that only high-quality, reliable data was used for further analysis and migration.

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6. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to uncover patterns, inconsistencies, and business insights.

Key analyses performed:

- Sales funnel analysis to track lead-to-cash conversion
- Pricing variance analysis to identify unauthorized discounts
- Revenue comparison between gross potential and realizable revenue
- Identification of high-risk customers based on financial exposure
- Trend analysis of sales and pipeline performance

This analysis helped in understanding the root causes of revenue leakage and operational inefficiencies.

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7. Tech Stack Used

- SQL: Data extraction, validation, joins, aggregations, and analysis
- Power BI: Dashboard creation, KPI tracking, and data visualization
- Excel: Data preprocessing and initial exploration
- ERP Concepts: Lead-to-Cash (L2C), Order-to-Cash (O2C), Record-to-Report (R2R)

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8. Key Findings

- 11.2% revenue leakage was identified due to pricing discrepancies and credit risks
- Unauthorized discounting contributed ~7.8% to total revenue loss
- High-risk credit exposure contributed ~3.4% potential financial loss
- Lack of validation controls in legacy systems was the primary root cause

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9. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Implement Data Validation Framework

- Use SQL-based validation checks before data migration
- Introduce automated data quality rules

2. Enforce Credit Risk Controls

- Apply dynamic credit limits (e.g., % of customer revenue)
- Automate order-blocking for high-risk transactions

3. Standardize Data Across Systems

- Maintain consistent field definitions across CRM and ERP
- Implement master data governance practices

4. Monitor Data Using Dashboards

- Use Power BI dashboards for real-time tracking of data quality and financial metrics
- Enable leadership visibility into key KPIs

5. Strengthen Data Governance

- Define clear ownership for data management
- Regular audits to prevent data inconsistencies

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10. Conclusion

This project highlights the importance of data analytics in ensuring successful ERP migration and financial accuracy. By combining data validation, exploratory analysis, and visualization, it was possible to uncover critical business issues and provide actionable insights.

The approach demonstrated in this project can be applied to real-world enterprise systems to improve data quality, reduce financial risk, and enable data-driven decision-making.

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